

LX / STAT: annotations



Date: 2026-10-31

notes on papers

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$LXstat
[1] "Stefan Th. Gries Statistics for Linguistics with R.pdf"
[2] "How_to_do_Linguistics_with_R._Data_exploration_and..._----_(1._What_is_statistics_).pdf"
[3] "How_to_do_Linguistics_with_R._Data_exploration_and..._----_(3._Descriptive_statistics_fo
[4] "How_to_do_Linguistics_with_R._Data_exploration_and..._----_(2._Introduction_to_R).pdf"


[1] "debug: number of annotations = 31"


docs annotated:


[1] "How_to_do_Linguistics_with_R._Data_exploration_and..._----_(1._What_is_statistics_).pdf"
[2] "How_to_do_Linguistics_with_R._Data_exploration_and..._----_(3._Descriptive_statistics_fo
[3] "Stefan Th. Gries Statistics for Linguistics with R.pdf"
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(16) annotations: How_to_do_Linguistics_with_R._Data_exploration_and..._----_(1._What_is_statistics_).p

paper: Levshina (2015)

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there is convenience sampling, the least reliable, but probably the most widely used method. One can simply make recordings of different speakers of Dutch only in a few easily accessible cities.

2

It is an ‘educated guess’ , which usually posits some difference between groups or relationship between variables

2

Representative sampling means that the researcher draws a sample in such a way that it matches the population on certain characteristics

2

The research hypothesis, which is also called, probably counterintuitively, the alternative hypothesis, always goes together with the null hypothesis

3

no difference between different groups, or no association between different variables

3

Therefore, you should not try to prove that your research (alternative) hypothesis is right. Instead, you should try to reject its opposite, that is, the null hypothesis

3

alternative hypotheses in A and B are called directional. They posit a direction to the inequality assumed by the researcher because they contain expressions like ‘X is more than Y’ and ‘the greater X, the greater Y’ . In contrast, the alternative hypothesis in C is non-directional. The researcher has no expectations about the frequencies of metaphoric expressions used by men and women. She simply expects to find a difference between the sexes. It can be expressed as ‘X does not equal Y

5

Probability distribution

5

Therefore, if one knows what distribution the data come from, it is possible to obtain the probability of observing the actual statistic (and more extreme values) by chance

6

If a p-value is smaller than some conventional level (usually 0.05 or 0.01), then the null hypothesis is rejected,

6

If the p-value is larger than this conventional value, then the null hypothesis cannot be rejected, and you can conclude that there is no sufficient evidence that the groups are different

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The outcome variable, or the one which changes as a function of some other parameters of interest, is called the response, or dependent variable

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The variables that influence the outcome are called explanatory, or independent variables.

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For example, consider the frequency of the word aardvark in a text.

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An example is answers in a questionnaire on a five-point Likert scale,

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If equal intervals on the scale represent equal differences between the points on the scale, we deal with an interval variable. A common example is temperature on the Celsius or Fahrenheit scale

(9) annotations: How_to_do_Linguistics_with_R._Data_exploration_and...—-(3._Descriptive_statistics_f

paper: Levshina (2015)

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The median is the value that is found in the middle of the ordered values.

4

median is a value below which we can find 50%. In other words, the median is the 50th percentile.

5

One can see that the maximum frequency is 16, which occurs twice: when the length is equal to eight and ten letters. The distribution therefore has two modes, eight and ten. Such distributions are called bimodal.

8

dispersion

verteilung. whats the difference to „distribution“?

8

variance is the sum of squared deviations of the observations from the mean divided by the number of observations minus 1

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The mean, the median and the mode cannot say anything about how much variation we have in our data. For this purpose, we need measures of dispersion

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median absolute deviation (MAD).

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Gries (2013: 120–125)

Gries (2021)

10

Gries (2013: 120–125) discusses such measures as relative entropy, average deviation and variation coefficient

(4) annotations: Stefan Th. Gries Statistics for Linguistics with R.pdf

paper: Gries (2021)

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Dispersion: normalized entropy

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evenness or relative diversity

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In other words, the higher Hnorm, the less informative the mode is

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This measure is the standard error (often abbreviated as se), which can in turn be used to compute something maybe even more important and widespread, a so-called confidence interval (often abbreviated as CI)

Gries, Stefan Th. 2021. *Statistics for Linguistics with R: A Practical Introduction*. De Gruyter. https://doi.org/10.1515/9783110718256.

Levshina, Natalia. 2015. *How to Do Linguistics with R: Data Exploration and Statistical Analysis*. Amsterdam: John Benjamins Publishing Company. https://doi.org/10.1075/z.195.

